

Boyan LI

✉ bli303@connect.hkust-gz.edu.cn

🌐 liboyan.vip

🔗 github.com/BugMaker-Boyan

🔍 Google Scholar

Education

Hong Kong University of Science and Technology (Guangzhou)

Guangzhou, China

Ph.D. Student in Data Science and Analytics (DSA)

Sep. 2024 – Present

Supervisor: **Prof. Yuyu LUO**

Southern University of Science and Technology (SUSTech)

Shenzhen, China

B.Eng. in Computer Science and Technology

Sep. 2019 – Jul. 2023

GPA: 3.91/4.0 (Rank 2/183) | Honors: Outstanding Graduate, Highest Honors

Publications

* indicates equal contribution. **Boyan Li** is bolded.

First-Author Publications (Including Co-First)

- [1] **DataSpace: Benchmarking Data Agents for Verifiable Analytics over Heterogeneous Workspaces**
Boyan Li*, Z. Liang*, Y. Xie*, X. Lin, T. Luo, X. Liu, Y. Zhu, Z. Peng, Y. Li, Z. Zhang, J. Zhang, N. Tang, G. Li, Y. Luo
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2027 (Under Review)
Official evaluation benchmark for the KDD Cup 2026 Data Agents for Complex Data Analysis competition.
- [2] **DeepEye-SQL: A Software-Engineering-Inspired Text-to-SQL Framework**
Boyan Li, C. Chen, Z. Xue, Y. Mei, Y. Luo
ACM SIGMOD International Conference on Management of Data (**SIGMOD**), 2026 (CCF-A)
- [3] **DeepEye: A Steerable Self-driving Data Agent System**
Boyan Li, Y. Peng, Y. Xie, S. Lu, Y. Zhu, X. Mu, X. Liu, Y. Luo
ACM SIGMOD International Conference on Management of Data (**SIGMOD**), 2026 (Demo, CCF-A)
- [4] **DPC: Training-Free Text-to-SQL Candidate Selection via Dual-Paradigm Consistency**
Boyan Li, O. O. K. Hei, Y. Yu, Y. Luo
Annual Meeting of the Association for Computational Linguistics (**ACL**), 2026 (Main Conference, CCF-A)
- [5] **ROSE: An Intent-Centered Evaluation Metric for NL2SQL**
W. Pei*, S. Hou*, **Boyan Li***, H. Chen, Z. Shi, Y. Luo
Annual Meeting of the Association for Computational Linguistics (**ACL**), 2026 (Main Conference, CCF-A)
- [6] **Alpha-SQL: Zero-Shot Text-to-SQL using Monte Carlo Tree Search**
Boyan Li, J. Zhang, J. Fan, Y. Xu, C. Chen, N. Tang, Y. Luo
International Conference on Machine Learning (**ICML**), 2025 (CCF-A)
Deployed in **Huawei Cloud DataArts Insight** and **Wuhan Urban Operation Agent**.
- [7] **DeepVIS: Bridging Natural Language and Data Visualization Through Step-wise Reasoning**
Z. Shuai*, **Boyan Li***, S. Yan, Y. Luo, W. Yang
IEEE VIS Conference (**VIS**), 2025 (CCF-A)
- [8] **The Dawn of Natural Language to SQL: Are We Fully Ready?**
Boyan Li, Y. Luo, C. Chai, G. Li, N. Tang
Proceedings of the VLDB Endowment (**VLDB**), 2024 (CCF-A)
VLDB 2024 Top-2 Highly Cited Paper (Google Scholar Metrics).

Co-authored Publications

- [9] **MarkovSQL: Adaptive Text-to-SQL Workflow Orchestration Under Partial Observability**
Y. Zhu, **Boyan Li**, Y. Luo
ACM SIGMOD International Conference on Management of Data (**SIGMOD**), 2027 (Under Review)
- [10] **SQLConductor: Search-to-Policy Learning for Step-wise Text-to-SQL Orchestration**
Y. Zhu, Z. Peng, **Boyan Li**, Y. Luo
Proceedings of the VLDB Endowment (**VLDB**), 2027 (Under Review)
- [11] **CoEvo-SQLi: Co-Evolving SQL Injection Detectors via Schema-Grounded Attack Synthesis**
H. Pan, X. Lin, **Boyan Li**, Y. Luo
International Conference on Learning Representations (**ICLR**), 2027 (Under Review)
- [12] **MIRA: Evidence-Verified Repair Memory for Text-to-SQL Correction**
Y. Liu, C. Yang, **Boyan Li**, R. Mao, Y. Luo
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (**KDD**), 2027 (Under Review)
- [13] **SCOPE-SQL: Structured Semantic Credit for Reinforcement Learning of Interactive SQL Repair Agent**
H. Qi, J. Zhao, **Boyan Li**, J. Jiang, Y. Luo
Conference on Empirical Methods in Natural Language Processing (**EMNLP**), 2026 (Findings)
- [14] **DataMagic: Transforming Tabular Data into Data Insight Video**
Y. Xie, C. Ma, Z. Wang, L. Wang, J. Zhu, C. Zeng, Z. Shen, **Boyan Li**, Y. Luo
Proceedings of the VLDB Endowment (**VLDB**), 2026 (Demo Track, CCF-A)

- [15] **How to Make Tool-Using LLM Agents Efficient? A Systematic Survey**
Y. Hu, Z. Cheng, W. Pei, M. Li, **Boyan Li**, Y. Zhu, Y. Xie, H. Chen, S. Hou, Y. Luo, H. Cheng, J. X. Yu, H. Du
Conference on Language Modeling (COLM), 2026 (Under Review)
- [16] **A Survey of NL2SQL with Large Language Models: Where are we, and where are we going?**
X. Liu, S. Shen, **Boyan Li**, P. Ma, R. Jiang, Y. Zhang, J. Fan, G. Li, N. Tang, Y. Luo
IEEE Transactions on Knowledge and Data Engineering (TKDE), 2025 (CCF-A)
- [17] **nvBench 2.0: A Benchmark for Natural Language to Visualization under Ambiguity**
T. Luo, C. Huang, L. Shen, **Boyan Li**, S. Shen, W. Zeng, N. Tang, Y. Luo
Conference on Neural Information Processing Systems (NeurIPS), 2025 (CCF-A)
- [18] **NL2SQL-BUGs: A Benchmark for Detecting Semantic Errors in NL2SQL Translation**
X. Liu, S. Shen, **Boyan Li**, N. Tang, Y. Luo
ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD), 2025 (CCF-A)
- [19] **Augmenting Realistic Charts with Virtual Overlays**
Y. Shi, **Boyan Li**, Y. Luo, L. Chen, N. Tang
CHI Conference on Human Factors in Computing Systems (CHI), 2025 (CCF-A)
- [20] **EllieSQL: Cost-Efficient Text-to-SQL with Complexity-Aware Routing**
Y. Zhu, R. Jiang, **Boyan Li**, N. Tang, Y. Luo
Conference on Language Modeling (COLM), 2025
- [21] **A Survey of Data Agents: Emerging Paradigm or Overstated Hype?**
Y. Zhu, ..., **Boyan Li**, et al.
IEEE Transactions on Knowledge and Data Engineering (TKDE), 2025 (Under Review)
[Ranked Top-3 in Hugging Face Daily Papers.](#)
- [22] **Advances and Challenges in Foundation Agents**
B. Liu, ..., **Boyan Li**, et al.
arXiv preprint, 2025
[Ranked Top-1 in Hugging Face Daily Papers.](#)
- [23] **Are Large Language Models Good Statisticians?**
Y. Zhu, S. Du, **Boyan Li**, Y. Luo, N. Tang
Conference on Neural Information Processing Systems (NeurIPS), 2024 (CCF-A)
- [24] **A Plug-and-Play Natural Language Rewriter for Natural Language to SQL**
P. Ma, **Boyan Li**, R. Jiang, J. Fan, N. Tang, Y. Luo
arXiv preprint, 2024

Open Source & Systems

- **DeepEye: A Steerable Self-driving Data Agent System** [\[GitHub\]](#)
– Led the development of a low-code orchestration platform for transparent data analysis.
– **Awards:** “Best Open-source Project Award” at AI Agent 2025 Competition; Silver Medal at the 51st Geneva International Exhibition of Inventions.
- **Alpha-SQL (ICML 2025)** [\[GitHub\]](#)
– Official implementation of the Monte Carlo Tree Search based Text-to-SQL agent.
– **Impact:** Deployed in **Huawei Cloud DataArts Insight** and **Wuhan Urban Operation Agent**.
- **NL2SQL Handbook (TKDE 2025)** [\[GitHub\]](#)
– A comprehensive and continuously updated roadmap for NL2SQL researchers.
– **Impact:** Widely recognized as a seminal resource for tracking state-of-the-art advancements, with GitHub Stars 1.4k+.
- **NL2SQL360 Evaluation Framework (VLDB 2024)** [\[GitHub\]](#)
– A comprehensive testbed for fine-grained evaluation of NL2SQL solutions.
– **Recognition:** Associated with the VLDB 2024 Top-2 Highly Cited Paper.

Experience

Tsinghua University (Database Group)
Exchange Student (Mentor: Prof. Guoliang Li)
Huawei Central Research Institute
Research Intern

Beijing, China
Jun. 2025 – Dec. 2025
Shenzhen, China
Jul. 2022 – Sep. 2022

Honors & Awards

Merit Prize , INFH Research Excellence Award	<i>May. 2026</i>
Silver Medal , 51st Geneva International Exhibition of Inventions (DeepEye Data Agent System)	<i>Mar. 2026</i>
Silver Medal , 51st Geneva International Exhibition of Inventions (Low-Altitude Economy Ground-Air Collaborative Delivery Digital Twin Platform)	<i>Mar. 2026</i>
Research Excellence Awards , DSA Excellence Awards 2025	<i>Dec. 2025</i>
Runner-up, Best Project Award , DSAA 6000R, HKUST(GZ)	<i>Nov. 2025</i>
Best Open-source Project Award , AI Agent 2025 Competition	<i>Oct. 2025</i>
Merit Prize , DSA Excellent Research Award 2024	<i>Jan. 2025</i>
Best Project Award , DSAA 5021, HKUST(GZ)	<i>Dec. 2024</i>
Highest Honors in Computer Science & Engineering , SUSTech	<i>Jun. 2023</i>
Outstanding Graduate , SUSTech	<i>Jun. 2023</i>
Huawei Scholarship , Huawei Technologies	<i>May. 2022</i>
Excellent Student Scholarship , SUSTech	<i>2020, 2021, 2022</i>
National Encouragement Scholarship , Ministry of Education of China	<i>Apr. 2020</i>

Services

General Chair : Data Agents for Complex Data Analysis (KDD Cup 2026)	<i>2026</i>
Program Chair : Agentic Data Systems (VLDB 2026 Workshop)	<i>2026</i>
Conference Reviewer : Neural Information Processing Systems (NeurIPS); International Conference on Learning Representations (ICLR); ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD); Annual Meeting of the Association for Computational Linguistics (ACL); Conference on Empirical Methods in Natural Language Processing (EMNLP)	
Journal Reviewer : IEEE Transactions on Knowledge and Data Engineering (TKDE)	
Invited Talk : Huawei Squirrel Webinar, “DeepEye-SQL”	<i>Jan. 2026</i>

Teaching

Teaching Assistant , Future Technology Exploration and Innovation Camp, HKUST(GZ)	<i>Jul. 2026</i>
Teaching Assistant , C++ Programming (UFUG 2601), HKUST(GZ)	<i>Spring 2025–26</i>
Teaching Assistant , Table Representation Learning (DSAA 6000O), HKUST(GZ)	<i>Spring 2024–25</i>